SS2 (STUDENT NOTE)

Contact Information
Course Timeline
Grading System
- Weekly Grading Criteria (For Midterm & Process Evaluation)
  - 1. Task Completion - 40% (4 points)
  - 2. Reporting & Tracking - 30% (3 points)
  - 3. Attitude & Problem Solving - 30% (3 points)
- Final Project Requirements

Suggested Topics

Enrollment Key: `ss2it0205spring2026`

Contact Information

- **Instructor:** Vinh NT
- **Email:** vinhnt@hanu.edu.vn
- **Note when contacting:** The email title must follow this exact format: `SS2_[ClassCode]_[StudentID] - Problem` (e.g., `SS2_03_2303400XX - Question about RAG`). If the format is not followed, your email will be ignored. Please include your Full Name and the specific issue in the body of the email.

Course Timeline

- **Weeks 2-3:** Technology introduction & **Topic Selection**
- **Remaining Weeks:**
  - Group and topic registration (Topics must be approved or assigned).
  - Implementation and weekly progress reporting.
  - Final presentation and submission of report/code.

Grading System

- **Attendance:** 10%
- **Midterm:** 30% (Derived from weekly progress reports and tracking files)
- **Final Project:** 60%

Weekly Grading Criteria (For Midterm & Process Evaluation)

Each student is evaluated on a 10-point scale every week based on the following criteria:

1. Task Completion - 40% (4 points)

- **4 points:** Completed 100% of the planned tasks from the previous week on time and with good quality.
- **3 points:** Completed most of the tasks (70-90%), with minor errors or slight delays.
- **2 points:** Completed half of the tasks (50%), basic quality, requires significant revisions.
- **0-1 points:** Incomplete or very little completed (<30%), or no concrete work outcomes.

2. Reporting & Tracking - 30% (3 points)

- **Using Tracking Template File: Link** (Don't edit this file directly, copy it to your Google Drive, public with view permission and edit it there)
- **3 points:** The tracking file is fully updated, clearly detailing completed tasks, assignees, and next week's plan. Clear and coherent reporting during review sessions.
- **2 points:** Tracking file updated but lacks detail or next week's plan is unclear.
- **1 points:** Very late updates to the tracking file, or updates do not reflect actual progress, confused reporting.
- **0 points:** Did not update the tracking file and did not report weekly progress.

3. Attitude & Problem Solving - 30% (3 points)

- **3 points:** Fully attended review sessions, actively contributed ideas to the team, proactively handled bugs/difficult problems during the week.
- **2 points:** Attended review sessions, contributed but passive in finding solutions or heavily dependent on others.
- **1 points:** Absent from review session (excused) or attended but unaware of own responsibilities in the project.
- **0 points:** Unexcused absence, completely shifting workload to other members.

Final Project Requirements

- **Group Size:** 2 to 3 members per team.
- **Topics:** Open (Web applications, Research, etc.). See suggested topics below.
- **Technology Stack:** No limitations.
- **Deliverables:**
  - Project Report
  - Presentation
  - Source Code

Suggested Topics

1. **"AI Study Buddy" (RAG Focus)**
   - **Description:** An app allowing students to upload study materials (PDF, Docx) and chat with them. AI responds based solely on uploaded knowledge.
   - **Tech Focus:** Retrieval-Augmented Generation (RAG), Vector Embeddings, Vector Database (ChromaDB/FAISS).
   - **Suggested Stack:** FastAPI + Jinja2 + LangChain/LlamaIndex + Gemini API.
2. **"Smart Github Explorer" (MCP Focus)**
   - **Description:** A tool to read and explain source code from repositories. Users provide a repo link/folder, and the AI accesses it directly to

analyze.

- **Tech Focus:** Model Context Protocol (MCP) to connect Gemini with local file systems or Github API.
- **Suggested Stack:** FastAPI + Google OAuth2 + Simple MCP Server.

3. **"AI Career Mentor" (Function Calling Focus)**
   - **Description:** A website supporting CV writing and mock interviews. AI acts as a recruiter asking questions based on the uploaded CV.
   - **Tech Focus:** Gemini Function Calling to extract CV information and search job markets (e.g., via Serper API).
   - **Suggested Stack:** FastAPI + Google OAuth2 + Gemini SDK.
4. **"Personal Knowledge Base" (Hybrid RAG & Google Workspace)**
   - **Description:** A smart notebook integrating with Google Drive/Docs. AI accesses user's documents (via OAuth2) to answer questions and search past notes.
   - **Tech Focus:** Google OAuth2 Scopes, RAG.
   - **Suggested Stack:** FastAPI + Google API Client Library + Gemini API.